2.3 Sequences

Subject content		Guidance	Curriculum reference
A	Find the term-to-term rule of a sequence	$5, 9, 13, 17, 21, \ldots$	A7.2A
B	Find terms of a sequence using a given term-to-term rule	A sequence starts with 8 and has a term-to-term rule of -3 What are the first 5 terms of the sequence? What is the 10th term of the sequence?	A7.2A A7.2B
C	Recognise different types of sequence	$2, 4, 6, 8, 10, \ldots$ $2, 4, 8, 16, 32, \ldots$ $1, 4, 9, 16, 25, \ldots$ $1, 3, 6, 10, 15, \ldots$ $1, 1, 2, 3, 5, 8, \ldots$ $1, 8, 27, 64, 125, \ldots$	A7.2C

2.3 Sequences *continued*

Subject content		Guidance	Curriculum reference
D	Generate terms of a sequence using a position-to-term rule in context	The position-to-term rule of a sequence is: 'Double the position number then subtract 1' Find the first 5 terms of this sequence.	A7.2D
E	Use the nth term to generate a linear or quadratic sequence	Find the first 5 terms of the sequences: $5n + 7$ $4n^2$	A7.2E A9.4A
F	Find the nth term of an arithmetic sequence	Find the nth term of the sequence: $2, 9, 16, 23, 30, \dots$	N9.4B
G	Recognise and continue more complex geometric sequences	Find the next term in the sequence: $1, 2.5, 5, 8.5, 13, \dots$	A9.4C
H	Solve problems involving sequences	Ali writes an arithmetic sequence that starts with: $4, 7, 10, 13, \dots$ Bhavin writes an arithmetic sequence that starts with: $15, 23, 31, 39, \dots$ Ali says that 111 is in both of their sequences. Is he correct? You must show your working.	A7.2F A9.4D